

SECR2043 OPERATING SYSTEMS

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Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	Output
	<code>pwd</code>	<code>chew@secr2043:~\$ pwd</code> <code>/home/chew</code>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	<code>ls -a ~</code>	<code>chew@secr2043:~\$ ls -a ~</code> <code>.</code> <code>.local</code> <code>file2.txt</code> <code>practice.txt</code> <code>..</code> <code>.profile</code> <code>filelist.txt</code> <code>sample.txt</code> <code>.bash_logout</code> <code>.sudo_as_admin_successful</code> <code>hello.txt</code> <code>workspace</code> <code>.bashrc</code> <code>combined.txt</code> <code>linux-lab</code>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	<code>mkdir ~/os_lab</code>	No Output
4.	Change your current working directory to "os_lab".	
	Command	Output
	<code>cd ~/os_lab</code>	No Output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output
	<code>echo "Hello, world!" > hello.txt</code>	No Output

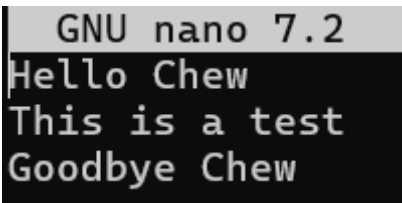
6.	Display the content of "hello.txt".	
	Command	Output
	<code>cat hello.txt</code>	<code>chew@secr2043:~/os_lab\$ cat hello.txt</code> Hello, world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	<code>cp hello.txt hello_copy.txt</code>	No Output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	<code>mv hello_copy.txt hello_bak.txt</code>	No Output
9.	Delete "hello.txt".	
	Command	Output
	<code>rm hello.txt</code>	No Output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	<code>ls -lR ~</code>	<code>chew@secr2043:~/os_lab\$ ls -lR ~</code> /home/chew: total 36 -rw-r--r-- 1 chew chew 158 Apr 16 13:53 combined.txt -rw-r--r-- 1 chew chew 79 Apr 16 13:48 file2.txt -rw-r--r-- 1 chew chew 364 Apr 16 13:57 filelist.txt -rw-r--r-- 1 chew chew 64 Apr 16 14:04 hello.txt drwxr-xr-x 2 chew chew 4096 Apr 16 14:41 linux-lab drwxr-xr-x 2 chew chew 4096 Apr 16 14:52 os_lab -rw-r--r-- 1 chew chew 718 Apr 16 14:16 practice.txt -rw-r--r-- 1 chew chew 814 Apr 16 14:14 sample.txt drwxr-xr-x 2 chew chew 4096 Apr 16 14:31 workspace /home/chew/linux-lab: total 0 /home/chew/os_lab: total 4 -rw-r--r-- 1 chew chew 14 Apr 16 14:52 hello_bak.txt /home/chew/workspace: total 44 -rwxr-xr-x 1 chew chew 16008 Apr 16 14:31 argument -rw-r--r-- 1 chew chew 398 Apr 16 14:31 argument.c -rwxr-xr-x 1 chew chew 15968 Apr 16 14:21 hello_world -rw-r--r-- 1 chew chew 76 Apr 16 14:18 hello_world.c -rw-r--r-- 1 chew chew 94 Apr 16 14:23 hello_world.cpp
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

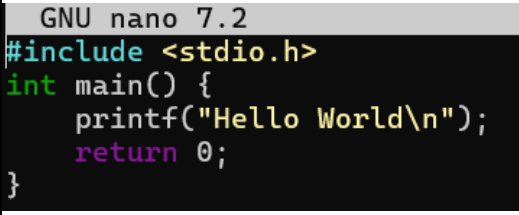
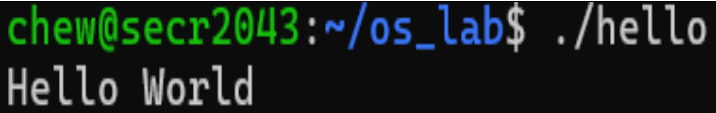
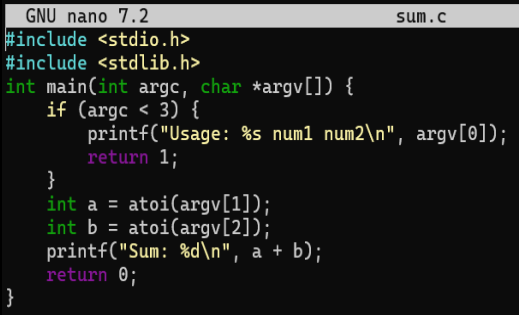
For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	<code>touch hello.txt</code>	No Output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	<code>echo "Hello World" > hello.txt</code>	No Output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	<code>echo "This is a test" >> hello.txt</code>	No Output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	<code>cat hello.txt</code>	<pre>chew@secr2043:~/os_lab\$ cat hello.txt Hello World This is a test</pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	<code>echo "Goodbye World" > bye.txt</code>	No Output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	<code>cat hello.txt bye.txt</code>	<pre>chew@secr2043:~/os_lab\$ cat hello.txt bye.txt Hello World This is a test Goodbye World</pre>

7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	<code>cat hello.txt bye.txt > greeting.txt</code>	No Output
8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	<code>cat greeting.txt</code>	<pre> chew@secr2043:~/os_lab\$ cat greeting.txt Hello World This is a test Goodbye World </pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	<code>nano greeting.txt</code> 	No Output
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	<code>cat greeting.txt</code>	<pre> chew@secr2043:~/os_lab\$ cat greeting.txt Hello Chew This is a test Goodbye Chew </pre>
Total Mark:		

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	<code>nano hello.c</code> 	No Output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	<code>gcc hello.c -o hello</code>	No Output
3.	Run the executable file "hello" and display its output.	
	Command	Output
	<code>./hello</code>	
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	<code>nano sum.c</code> 	No Output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	<code>gcc sum.c -o sum</code>	No Output

6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	<code>./sum 10 20</code>	<code>chew@secr2043:~/os_lab\$./sum 10 20 Sum: 30</code>
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	<code>./sum -5 15</code>	<code>chew@secr2043:~/os_lab\$./sum -5 15 Sum: 10</code>
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	<code>nano echo.c</code>	No Output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	<code>gcc echo.c -o echo</code>	No Output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	<code>./echo</code>	<code>chew@secr2043:~/os_lab\$./echo Enter text: Hello World The text entered is: Hello World</code>
Total Mark:		